ENTERPRISE FRAMEWORK

AI-Augmented SDLC Process

A complete guide to integrating AI tools across every phase of the software development lifecycle

AI-POWERED DELIVERY | 6-PHASE FRAMEWORK | TOOLS MAPPING

THE 6-PHASE AI-SDLC OVERVIEW

AI tools integrated at every stage to accelerate delivery and reduce errors

01 Planning

Requirements, risk
& sprint strategy

02 Design

Architecture, UX
& API contracts

03 Development

AI pair coding
& implementation

04 Testing

Auto test gen
& QA automation

05 Deployment

CI/CD pipelines
& IaC automation

06 Monitoring

Observability
& incident AI

AI tools reduce overall delivery cycle time by 30–55% across all phases

01 Planning & Strategy

Define scope, requirements, and project roadmap with AI-assisted analysis and risk detection.

⚡ AI IMPACT

AI reduces requirements definition time by ~40% through automated gap analysis and ambiguity detection.

KEY ACTIVITIES

- Stakeholder interviews & requirements
- Feasibility analysis & risk register
- Sprint planning & backlog creation
- Architecture decision records (ADRs)

TOOLS & AI INTEGRATIONS

Claude (Anthropic)
AI ASSISTANT

Requirements analysis, ADR drafting, user story generation

Jira
PROJECT MGMT

Backlog management, sprint planning, issue tracking

Confluence / Notion AI
DOCUMENTATION

Meeting summaries, project wiki, technical specs

Miro
DIAGRAMMING

System architecture, user journey mapping

GitHub Copilot
AI CODING

Prototype scaffolding, PoC code generation

01 Planning & Strategy — Examples in Action

Real prompts & AI outputs your team can use today

User Story Generation

CLAUDE

INPUT

Prompt to Claude:
"We need a checkout flow for our e-commerce app. Users should be able to pay."

▼

AI OUTPUT

✓ 12 user stories generated:
- Happy path purchase flow
- Payment failure & retry
- Guest checkout (no login)
- Cart abandonment recovery
- Accessibility (screen reader)
- Mobile layout edge cases
- Saved payment methods
- Order confirmation email

...+ 4 more with acceptance criteria

Sprint Risk Flagging

CLAUDE

INPUT

Prompt to Claude:
"Our team of 4 devs will build a multi-tenant SaaS app with SSO and billing in 3 months."

▼

AI OUTPUT

⚠️ Risks identified:
- Tenant data isolation = 3+ weeks
- SSO integration (SAML/OAuth) = 2 weeks
- Billing (Stripe) edge cases = 1.5 weeks

→ Recommendation: Phase the MVP.
Month 1: Core app + single tenant
Month 2: Multi-tenancy
Month 3: SSO + billing

02 Design & Architecture

System design, UX/UI, and technical architecture with AI-generated patterns and design reviews.

⚡ AI IMPACT

AI generates 60% of boilerplate architecture patterns and auto-identifies design anti-patterns.

KEY ACTIVITIES

- System architecture & API design
- Database schema design
- UI/UX wireframes & prototypes
- Security threat modeling

TOOLS & AI INTEGRATIONS

Claude (Anthropic)
AI ASSISTANT

Architecture review, schema & OpenAPI spec generation

Figma AI
DESIGN

UI component generation, design system enforcement

GitHub Copilot
AI CODING

Boilerplate scaffolding, pattern suggestions

draw.io / Miro
DIAGRAMMING

ERDs, sequence diagrams, service maps

Postman
API DESIGN

API mocking, contract testing, documentation

02 Design & Architecture — Examples in Action

Real prompts & AI outputs your team can use today

API Spec from Schema

CLAUDE

INPUT

Prompt:
"Given this DB schema (users, orders, products tables), generate a REST API spec for order management."

▼

AI OUTPUT

✓ Full OpenAPI 3.0 spec generated:

GET /orders → list + pagination
POST /orders → create, validates stock
GET /orders/{id} → detail + line items
PATCH /orders/{id} → status update
DELETE /orders/{id} → soft delete

Includes: error codes, auth headers, request/response schemas, examples

Architecture Risk Review

CLAUDE

INPUT

Prompt:
"We plan to use a monolith with a shared Postgres DB for all microservices to save cost."

▼

AI OUTPUT

⚠️ 3 anti-patterns detected:

1. Shared DB = tight coupling, blocks independent deployments
2. Schema conflicts risk data corruption
3. Scaling bottleneck at DB layer

→ Suggested: Database-per-service pattern with async events (Kafka) for cross-service communication

03

Development

AI-augmented coding with intelligent pair programming, auto-completion, and code generation.

⚡ AI IMPACT

Developers report 35–55% productivity gains with AI-assisted coding on routine and boilerplate tasks.

KEY ACTIVITIES

- Feature development & implementation
- AI pair programming sessions
- Code refactoring & optimization
- Component library development

TOOLS & AI INTEGRATIONS

GitHub Copilot
AI CODING

Inline completion, function gen, docstring writing

Claude (Anthropic)
AI ASSISTANT

Debugging, code review, complex logic, refactoring

Cursor IDE
AI IDE

Codebase-aware AI with multi-file editing

VS Code + Extensions
IDE

Primary development environment, linting, formatting

Docker
CONTAINERS

Consistent dev environments, service isolation

03 Development — Examples in Action

Real prompts & AI outputs your team can use today

Bug Root-Cause from Stack Trace

CLAUDE

INPUT

Paste cryptic error to Claude:
"TypeError: Cannot read properties
of undefined (reading 'price')
at checkout.js:142
at processCart()"

AI OUTPUT

✓ Root cause identified:
The cart item object returned by
getCartItems() can be null when
a product is deleted mid-session.

Line 142 doesn't null-check before
accessing item.price.

Fix: Add optional chaining:
const price = item?.price ?? 0;

Refactor to Repository Pattern

CURSOR IDE

INPUT

Prompt in Cursor:
"Refactor this 280-line UserController into the repository pattern. Keep tests green."

AI OUTPUT

✓ 4 files created/updated:
- UserRepository.js (data layer)
- UserService.js (business logic)
- UserController.js (slim, 45 lines)
- UserRepository.test.js (new tests)

All 23 existing tests still passing.
Cyclomatic complexity: 18 → 6
Test coverage: 71% → 89%

04

Testing & QA

AI-generated test cases, automated regression testing, and intelligent bug triage at scale.

⚡ AI IMPACT

AI generates test coverage boilerplate 70% faster and surfaces edge cases humans routinely miss.

KEY ACTIVITIES

- Unit & integration test authoring
- End-to-end test automation
- Performance & load testing
- Security vulnerability scanning

TOOLS & AI INTEGRATIONS

Claude (Anthropic) — AI ASSISTANT

Test case generation, edge-case discovery, bug RCA

GitHub Copilot — AI CODING

Auto-generate unit tests from function signatures

Playwright — E2E TESTING

Browser automation, UI regression testing

Jest / Vitest — TESTING

Unit & integration test runner

Snyk — SECURITY

AI-powered vulnerability detection in dependencies

04 Testing & QA — Examples in Action

Real prompts & AI outputs your team can use today

Unit Test Generation

COPILOT + CLAUDE

INPUT

Paste function signature to AI:

```
function processPayment(
  amount, currency, cardToken,
  idempotencyKey
) { ... }
```

AI OUTPUT

✓ 14 tests generated in Jest:

- Happy path (USD, valid card)
- Declined card → PaymentError
- Network timeout → retry logic
- Duplicate idempotency key
- Zero amount → validation error
- Negative amount → rejection
- Unsupported currency code
- Expired card token
- Missing required fields (x4)

E2E Checkout Test (Playwright)

CLAUDE

INPUT

Prompt:

"Write a Playwright test for the full checkout: login → add item → pay → confirm order."

AI OUTPUT

✓ Complete test script generated:

```
test('checkout flow', async ({page}) => {
  await page.goto('/login');
  await page.fill('#email', user);
  await page.fill('#pass', pass);
  await page.click('#login-btn');
  await page.click('#add-to-cart');
  await page.click('#checkout');
  await page.fill('#card', '4242...');
  await page.click('#pay-now');
```

05 Deployment & CI/CD

Automated pipelines with AI-assisted config generation, IaC authoring, and anomaly detection.

⚡ AI IMPACT

AI-generated IaC templates reduce misconfiguration errors by ~50% and cut pipeline setup time in half.

KEY ACTIVITIES

- CI/CD pipeline configuration
- Infrastructure as Code (IaC)
- Release management & feature flags
- Deployment rollback strategies

TOOLS & AI INTEGRATIONS

Claude (Anthropic)
AI ASSISTANT

Dockerfile & pipeline YAML generation, IaC review

GitHub Actions
CI/CD

Automated build, test, and deployment pipelines

Terraform
IAC

Cloud infrastructure provisioning and state management

AWS / GCP / Azure
CLOUD

Hosting, managed services, auto-scaling

LaunchDarkly
FEATURE FLAGS

Gradual rollouts, A/B testing, kill switches

05 Deployment & CI/CD — Examples in Action

Real prompts & AI outputs your team can use today

GitHub Actions Pipeline

CLAUDE

INPUT

Prompt:
"Generate a GitHub Actions workflow to deploy a Node.js app to AWS ECS with staging gate and Slack notification."

AI OUTPUT

✓ deploy.yml generated (68 lines):

- Trigger: push to main branch
- Stage 1: npm install + Jest tests
- Stage 2: Docker build + ECR push
- Stage 3: Deploy to ECS (staging)
- Gate: manual approval required
- Stage 4: Deploy to ECS (prod)
- On failure: auto-rollback task
- Notify: Slack #deployments channel

Terraform Multi-AZ RDS

CLAUDE

INPUT

Prompt:
"Write Terraform for a production Postgres RDS instance: multi-AZ, read replica, automated backups, encrypted."

AI OUTPUT

✓ main.tf generated:

```
resource "aws_db_instance" "prod" {
  engine        = "postgres"
  instance_class = "db.r6g.large"
  multi_az      = true
  storage_encrypted = true
  backup_retention = 7
  deletion_protection = true
}
resource "aws_db_instance" "replica"
```

06 Monitoring & Operations

Intelligent observability, AI-driven incident response, and continuous feedback loops.

⚡ AI IMPACT

AI-driven anomaly detection reduces mean time to detect (MTTD) by up to 60% vs. manual monitoring.

KEY ACTIVITIES

- Real-time performance monitoring
- AI-assisted incident triage
- Log analysis & anomaly detection
- Retrospectives & continuous improvement

TOOLS & AI INTEGRATIONS

Claude (Anthropic)
AI ASSISTANT

Log analysis, post-mortems, runbook authoring

Datadog
OBSERVABILITY

Metrics, traces, logs with AI anomaly detection

PagerDuty AI
INCIDENT MGMT

On-call management, intelligent alert triage

Sentry
ERROR TRACKING

Frontend/backend error capture and grouping

Grafana
DASHBOARDS

Custom metrics dashboards and alerting rules

06

Monitoring & Operations — Examples in Action

Real prompts & AI outputs your team can use today

Incident Diagnosis from Logs

CLAUDE

INPUT

Paste Datadog spike + logs to Claude at 2am:
"DB connection errors spiking.
500 errors up 800% last 10 min.
Here are the last 50 log lines..."

AI OUTPUT

✓ Root cause in 30 seconds:
Connection pool exhausted.
Query on orders table doing
full table scan (missing index).

Triggered by: new feature deployed
at 1:47am — uses ORDER BY
created_at without index.

Immediate fix:

AI-Generated Post-Mortem

CLAUDE

INPUT

Paste incident timeline + Slack thread to Claude:
"S3 latency spike → 47 alerts → 22 min outage. Here is the full timeline..."

AI OUTPUT

✓ Blameless post-mortem generated:
Impact: 22 min, ~4,200 users
Root cause: S3 us-east-1 latency
Contributing: no circuit breaker

5 Whys analysis included.

Action items (8 total):
1. Add S3 circuit breaker [Eng, 1w]
2. Multi-region S3 fallback [Eng, 2w]

AI TOOLS MATRIX

Where each AI tool applies across the SDLC

Tool	Plan	Design	Dev	Test	Deploy	Monitor
Claude (Anthropic)	✓	✓	✓	✓	✓	✓
GitHub Copilot	—	✓	✓	✓	—	—
Cursor IDE	—	—	✓	—	—	—
Figma AI	—	✓	—	—	—	—
Jira / Confluence	✓	✓	—	—	—	—
GitHub Actions	—	—	—	✓	✓	—
Datadog / Sentry	—	—	—	—	—	✓
Snyk	—	—	—	✓	✓	—
Terraform	—	✓	—	—	✓	—
PagerDuty AI	—	—	—	—	—	✓

Measurable AI Impact

Key productivity and quality gains across the AI-SDLC

40% Faster requirements definition (Planning)

55% Developer productivity gain (Development)

70% Faster test case generation (Testing)

60% Reduction in MTTD (Monitoring)

The AI-SDLC is not the future — it is the competitive standard today.